# Closing Modality Gap of Multi-vector Visual Document Retrieval

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## 1 Topic Statement

Universal multimodal embedding models have achieved great success in capturing semantic relevance between queries and candidates. However, current methods either condense queries and candidates into a single vector, which may limit the expressiveness for long-context documents (Faysse et al., 2024). On the other hand, multi-vector multimodal retrieval enhances the accuracy for long and cross-document queries with multiple vector representations and late interaction. Yet, the performance on structure-sensitive elements like tables and charts remains limited. In this proposal, we aim to enhance the current multi-vector retrieval paradigm for chart-rich documents by designing effective late interaction methods.

**Single-vector Multimodal Retrieval.** Following the proposal of CLIP, single-vector multimodal embedding has been extensively explored, where the modality gap is a fundamental challenge. Specifically, while contrastive learning attempts to align text and image embeddings in the same latent space, it inadvertently facilitates the formation of a modality gap (Li et al., 2025), where embeddings of the same modality tend to cluster together, thus degrading the performance of cross-modal alignment, displayed in Figure 1. Moreover, the context length limitations of base models often constrain their application in long document retrieval.

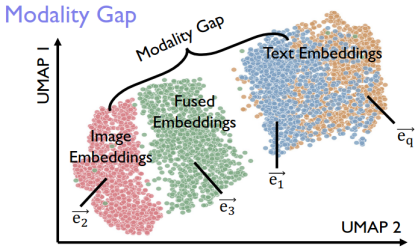


Figure 1: Illustration of Modality Gap in Multimodal Embedding Models.

**Multi-vector Multimodal Retrieval.** To address the challenge of document-level retrieval, traditional methods often extract text from multimodal documents using techniques such as OCR and layout detection, followed by a conventional chunking-then-retrieving pipeline with text embeddings (Faysse et al., 2024). Such methods are not only complicated but also inefficient for documents rich in visual elements. The emergence of multimodal large language models (MLLMs), however, provides more effective solution for multimodal document retrieval by outputting aligned visual embeddings in text space (Meng et al., 2025), with late interaction applied to retrieval related candidate accurately, as shown in Figure 2. Despite these efforts, current multi-vector retriever still struggles with structured elements like charts and tables (Xu et al., 2025).

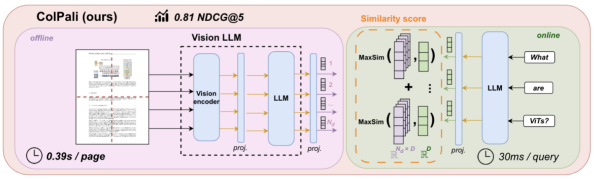


Figure 2: Illustration of multi-vector retrieval with MLLM.

## 2 Literature Review

**Theoretical Analysis of Embedding Models.** Initially formalized by Voronoi (Voronoi), the Voronoi diagram, analogous to information retrieval, has been studied in mathematics for many years. Weller et al. (2025) further extends this theory in neural embedding models,

discussing the limitations of single-vector retrieval. On the other hand, the theory of modality gap and its mitigation has also been extensively studied. For example, Li et al. (2025) identifies the formation of the modality gap, which stems from the contrastive training process, while Role et al. (2025) and Cicchetti et al. (2024) discuss post-hoc transformations to close the gap in the embedding space. There is also research attempting to decipher multimodal embedding using advanced toolkits like Algebraic Geometry (Zhao, 2024), although practical insights for real-world retrieval tasks are still lacking.

**Multi-vector Retrieval Baselines.** Initially proposed by Colpali (Faysse et al., 2024), multi-vector retrieval models have achieved state-of-the-art performance in multimodal document retrieval tasks. Among them, jina-embeddings-v4 (Günther et al., 2025) incorporates task-specific LoRA adapters based on Qwen2.5-VL-3B-Instruct tasks, while VLM2Vec-V2 (Meng et al., 2025) proposes a unified embedding model for video, image, and visual documents based on the Qwen2-VL backbone. Llama Nemoretriever Colembed (Xu et al., 2025) advances multimodal document retrieval by large-scale pre-training on text-only retrieval corpora using a contrastive loss. Despite these efforts, methods for documents rich in charts and tables still underperform.

## 3 Methodology and Contributions

**Survey on Multimodal Embedding Research.** This section investigates the current theoretical and practical research on multimodal embedding, with particular emphasis on the modality gap and its mitigation. The analysis and insights into chart-related document retrieval should also be highlighted, with lightweight late interaction preferred to enable plug-and-play enhancement of table-rich document retrieval.

**Methodology Formulation.** The methodology for this project will be theory-inspired. Specifically, the methodology will include the following parts:

- **Theoretical Analysis and Pilot Experiment.** A theoretical analysis of the structure of the latent space for current multimodal embedding models, with a focus on the modality gap and chart element representation.
- **Theory-Inspired Late Interaction.** An enhanced design for the late interaction algorithm will be proposed, aiming to facilitate the localization of structured visual elements like charts and tables.
- **Downstream Generalization.** Extending the generalizability and scalability of the proposed method, ensuring that performance can be continuously improved in downstream and cross-domain tasks in a robust manner.

**Evaluation on Domain-Specific Scenarios.** The evaluation should consider the following aspects, which are critical for real-world deployment:

- **Performance.** The accuracy and precision of the retrieval method should be comprehensively considered, with common metrics such as nDCG@5 being continuously improved.
- **Computation Consumption.** While task performance in document retrieval is emphasized, computation consumption is also critical for real-world applications. A comprehensive analysis of the tradeoff between performance and computational consumption is required.
- **Robustness.** Sensitivity analysis of possible hyperparameters and experimental settings will be conducted to ensure robust improvements rather than results influenced by overfitting or fine-tuning.

## 4 List of Courses

- *MATH 549. Differentiable Manifolds I*. Smooth manifolds and maps, tangent and normal bundles, Sard's theorem and transversality, embedding, differential forms, Stokes's theorem, degree theory, vector fields.
- *CS 582. Information Retrieval.* Document retrieval, office automation. Optimal retrieval, relevance feedback, clustered search, construction of clusters, model of term weighting, thesaurus construction, multimedia data, handling of audio and video.
- *CS 477. Public Policy, Legal, and Ethical Issues in Computing, Privacy, and Security.* Contemporary topics involving legal, public policy, and/or ethical issues in computing, especially privacy, security, and surveillance.

## 5 Timeline

The proposed project will be conducted over a period of three months. The timeline is divided into four distinct phases, each lasting approximately three weeks, to ensure a systematic progression from theoretical analysis to final evaluation.

- **Phase 1: Theoretical Analysis and Pilot Experiments (Weeks 1–3)**
  - Conduct a comprehensive survey on the latest multimodal embedding research, focusing on the modality gap and chart-element representation.
  - Perform a theoretical analysis of the latent space structure in current embedding models, drawing upon Voronoi diagram theories.
  - Execute pilot experiments to quantify the modality gap and identify specific failure cases in retrieving structured elements like charts and tables.
- **Phase 2: Algorithm Design and Late Interaction Development (Weeks 4–6)**
  - Formulate the Theory-Inspired Late Interaction mechanism aimed at improving the localization of structured visual elements.
  - Design lightweight interaction modules that facilitate plug-and-play enhancements for table-rich documents without incurring excessive computational overhead.
  - Develop the initial prototype of the improved multi-vector retrieval model.
- **Phase 3: Implementation and Downstream Generalization (Weeks 7–9)**
  - Implement the full retrieval pipeline integrating the proposed late interaction method.
  - Conduct experiments to extend the generalizability of the method across different domains.
  - Optimize the model to ensure scalability and robustness in downstream cross-domain tasks.
- **Phase 4: Comprehensive Evaluation and Analysis (Weeks 10–12)**
  - Evaluate the model's performance using standard metrics (e.g., nDCG@5) on chart-rich document datasets.
  - Perform a trade-off analysis between retrieval accuracy and computation consumption.
  - Conduct sensitivity analysis on hyperparameters and experimental settings to verify the robustness of the improvements and rule out overfitting.

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